Ashwathi Nair

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Summary

I am a theoretical and computational astrophysicist with expertise and interest in stellar and binary star formation and evolution, their observations, numerical simulations, and statistical analysis of complex datasets. Experienced in building and designing efficient, robust, and reproducible open-source computational workflows. Strong independent learner and problem-solver with strong project management skills and a commitment(advocating) to(for) inclusive and accessible research and workplace practices.

Skills

- **Scientific codes:** Binary Population Synthesis Codes: MESA, COMPAS, POSYDON, binary-c, CLOUDY, PHANTOM, Mathematica
- **Programming language:** Python, C++
- **Data & Analysis:** NumPy, Matplotlib, SciPy, Pandas, AstroPy, GWpy, Astrostatistics
- **High Performance Computing:** Experience with GPU (Slurm, batch jobs)
- **Project Management & Communication:** Research workflow management, project milestones establishments

Education

Astrophysics Ph.D. Candidate — Melbourne, Australia
Centre for Astrophysics and Supercomputing, Swinburne University of Technology & OzGrav — 2023–Ongoing

- Thesis: "Discovering the Origins of Gravitational Wave Sources"
  Advisors: Dr. Simon Stevenson and Prof. Jarrod Hurley

Master in Science (Physics - Astrophysics) — Mumbai, India
St. Xavier's College — 2019–2021

- Thesis: "Shadow of Ultra-high Density Compact Objects"
  Advisors: Dr. Dipanjan Dey and Prof. Pankaj S. Joshi

Bachelor in Science (Physics) — Mumbai, India
K.J. Somiaya College of Science and Commerce — 2013–2018

Scholarships and Awards

- SRASS (Australian Research Council (ARC) funded scholarship) — ∼ AUD 120000
- Travel Funds (domestic and international) — ∼ AUD 15000
- Charotar University of Science and Technology (CHARUSAT) Master's Thesis stipend award — ∼ INR 63000

Selected Research Experience

PhD research: Discovering the Origins of Gravitational Wave Sources — Melbourne, Australia
Swinburne University of Technology, OzGrav — 2023 - Ongoing

- Developed extensive numerical simulations of single and binary stellar evolution using Modules of Experiments in Stellar Astrophysics (MESA) to study formation pathways of double neutron star and black hole-neutron star systems.
- Designed and built a post-processing pipeline to perform population synthesis, including different binary phases, modelling of supernova remnants, orbital evolution, and statistical weighting of simulated grids.
- Performed statistical analysis of high-dimensional parameter spaces to identify key formation pathways.

- Planned, designed, and worked with an end-to-end computational workflow resulting in two first-authored papers, along with the third in preparation

Master's research: Shadow of Ultra-high Density Compact Objects Gujarat & Mumbai, India
St. Xavier's College and Charotar University of Science and Technology 2021

- Modeled photon trajectories in curved spacetime by applying the Hamilton–Jacobi formalism using Mathematica (Simulated light trajectories and the shadow of compact objects using the back-ray tracing method). Analysed conditions for shadow formation in both singular and regular black-hole spacetimes.
- Investigated the relationship between the shadow of a compact object (spherically symmetric and static) and a space-time singularity.
- Theoretically worked out the conditions for the presence and absence of the photon sphere, and conditions for which shadow could be seen in an observer's sky.
- Extended our study to understand what other possible ways in which light could get trapped, by studying regular black hole spacetimes.

Spin Parameter Estimation of Compact Object from its Shadow Nature Gujarat, India
Independent Project 2022-2023

- Developed numerical simulations to quantify relationships between shadow observables and compact-object spin parameters.

A Study of Massive Highly Rotating Stars (Wolf-Rayet Stars) Mumbai, India
Advisor: Gururaj Wagle, St. Xavier's College 2020

- Studied the evolution of massive stars with different rotational velocities and metallicities towards Wolf-Rayet stars using the detailed stellar evolution code MESA.

Publications

- **Nair, Ashwathi**, Stevenson, Simon, 2025, "Formation of heavy double neutron stars I. Eddington-limited accretion for a 1.4 $M_\odot$ neutron star at solar metallicity", MNRAS, 543, 233–246
- **Nair, Ashwathi**, Stevenson, Simon, 2026, "Formation of heavy double neutron stars II: the role of heavy first-born neutron stars and low metallicity", arXiv:2608.17430
- **Nair, Ashwathi**, Stevenson, Simon, "Formation of heavy NSs in black hole-neutron star binaries"– In prep.

The following are a few selected ongoing collaborations.

- Stevenson, Simon, **Nair, Ashwathi**, Final Authors TBD, "Binary evolution predicts heavier neutron stars in mixed compact binaries " – In prep.
- Simon Stevenson, Reinhold Willcox, **Ashwathi Nair**, and Alejandro Vigna-G'omez "PSR J0210+5845 constrains the occurrence of electron-capture supernovae in super-asymptotic giant branch stars " – To be submitted in December 2026.
- E. Carli, Y. Song, S. Stevenson, K. Auchettl, L. Levin, **A. Nair**, B. W. Stappers, J. D. Turner, M. Bailes, C. Flynn, R. M. Shannon, R. P. Breton, F. Calore, E. D. Barr, M. Kramer, "The TRAPUM Small Magellanic Cloud pulsar survey with MeerKAT: III. The neutron star population of the SMC" – To be submitted in December. Contributed to the sections on binary population synthesis and mainly to the sections on interpreting the role of metallicity on binary interaction, supernova physics, kicks and the NS population properties in the SMC.
- Alan R. Duffy, Chris Flynn, Kenneth Freeman, Renée Key, Anais Möller, Jeremy Mould, **Ashwathi Nair**, Abhijit Saha, and E.N. Taylor, "Variable stars in the LMC. I: Two short-period contact binaries, one pulsating white dwarf and a host of δ Scuti stars" – In prep.
  Contributed to determining the formation of the observed short-orbital-period binaries (potential double degenerate binaries).

Selected Talks

Seminars and Invited Talks

- Visiting Seminar, Institute of Science and Technology Austria (ISTA) — Austria, 2026
  Visitor at Ylva Götberg's group
- Visiting Seminar, Waseda University — Japan, 2026
  Visitor at Shoichi Yamada's lab
- Invited Panelist, OzGRAV-KAGRA Summer School/Collaboration — Japan, 2026
  To lead an advanced discussion on gravitational wave progenitors and population
- Invited Talk, COMPAS code meeting, Monash University — Australia, 2025
- Invited Seminar, University of Melbourne — Australia, 2024

Contributed Talks

- Departmental Review talk, Centre for Astrophysics and Supercomputing, Swinburne University of Technology — Australia, 2024, 2025, 2026
- Contributed talk, Binary Stars in the New Era Conference — China, 2025
- Contributed talk, GR24 & Amaldi16 International Conference — Glasgow, 2025
- Contributed talk, Australian National Institute for Theoretical Astrophysics (ANITA) Conference — Australia, 2025
- Contributed Sparkler talk, OzGrav Retreat and Conference — Australia, 2024, 2025

Professional Training

Selected Schools and Workshops

- **POpulation SYnthesis with Detailed binary-evolution simulatiONs (POSYDON) Summer School** — Switzerland 2026
  Intensive four-day school at the Geneva Observatory spent learning how to use POSYDON in depth
- **International CEntre of SUperNovae (ICESUN) Summer School 2025: Binary Stars and Compact Objects** — China 2025
  binary-c population synthesis code, spectroscopic observational techniques.
- **MESA Down Under Winter School and Workshop** — Australia, 2024
- **Phantom SPH code at ANITA Workshop** — Australia 2024
  Learned how to work with smoothed particle hydrodynamics (SPH) simulation with the Phantom SPH code.
- **HPC and Data in Astrophysics** — Online, Australia 2024
  This course was designed for training and supporting Australian and international HDR students and ECRs working on complex astrophysics problems requiring HPC and data tools. The course covered diverse aspects of computational astrophysics, such as numerical simulations; HPC and data techniques; statistical modelling and inference; and offered hands-on tutorials.
- **OzGrav ECR Workshop.** — Australia 2023, 2024, 2025
  Training on a broad range of topics: Hierarchical Bayesian Inference, Data Analysis, Astrostatistics, etc.
- **Summer School on Gravitational-Wave Astronomy** — Bengaluru, India 2022
  Training on formation channels, supernova explosion physics at the International Centre for Theoretical Sciences of the Tata Institute of Fundamental Research (ICTS-TIFR)

Selected Conferences

- **GR24/Amaldi16 2025** — Scotland 2025
  The 24th International Conference on General Relativity and Gravitation (GR24) and the 16th Edoardo Amaldi Conference on Gravitational Waves (Amaldi16)
- **Binary Stars in a New Era** — China 2025
  Yunnan Observatories, Chinese Academy of Sciences (CAS)

- **Australian National Institute for Theoretical Astrophysics (ANITA) Conference** — Australia, 2024, 2025
- **Astronomical Society of Australia Conference** — Australia, 2024, 2025
- **Transient Down Under** — Australia, 2024
- **LISA SYMPOSIUM XIV** — Hybrid, 2022,

Professional Service and Leadership

Community Service

- Peer reviewer for **The Astrophysical Journal** (ApJ) (x1) — 2026
- Student Representative of the Equity, Diversity and Inclusion Committee at the Centre of Astrophysics and Supercomputing, Swinburne University of Technology. Contributing to initiatives supporting diversity and inclusion, including developing a support system for neurodiverse individuals. — 2023–Present
- Chair for a panel discussing career pathways at OzGrav Retreat — 2025
- OzGrav ECR Committee Member and ECR Workshop Organizer — 2024-2025
- Team Member at Include Her, promoting awareness and representation of women scientists and their contributions in school curricula. — 2024–2025
- Women in Physics Breakfast — 2024–2025

Teaching and Mentoring

- Designed and delivered an introductory stellar astrophysics course for Year 10–12 short-term work-experience students, including hands-on coding projects and supervised mentoring on their research and communication skills. — Australia, 2024–2025
- Science Professional Mentor — India & Australia, 2023–Present
  1. Mentor to Honors student Ella Cross currently at RMIT, Melbourne.
  2. Provided end-to-end support to five Master's students from St. Xavier's College applying for PhD positions by instructing them on their talks, shaping their research statements, and preparing for interviews, leading to successful placements.
- High School Physics and Mathematics Teacher — India, 2018–2019

Outreach and Leadership

- **AstroEdu Conference** — Australia, 2025
  Presented a talk discussing the value of bringing astronomy education to schools and how this fundamentally helps PhD researchers in communicating their research and personal development. The conference was a rich experience that brought together astronomy educators from all around the world. This provided a great experience in learning the tools and methods used to make astronomy accessible to non-experts, experts, and everyone in between.
- Conducted outreach at 10+ Victorian high schools and led four of those sessions as an Outreach Ambassador for OzGrav — Australia
- Represented Outreach done at Swinburne University of Technology and OzGrav at the Victorian Parliament — Australia, 2025
- Designed astronomy content for astronomy labs and schools established in rural areas of India for students aged 6–15, enhancing active participation and engagement of students. The materials were also translated in different Indian languages — India, 2022
- Experience enabler and telescope manager at Stargazing Mumbai — India, 2019-2021

Media

Gravity's Echo – Exploring Gravitational Waves
An interview I did in 2024 for the Cosmic Web magazine, an initiative by the National Space Society (USA) - Mumbai, India Edition.

References

Dr. Simon Stevenson
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Prof. Jarrod Hurley
PhD Co-Supervisor
Swinburne University of Technology
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